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FROM GREY TO GREEN

WITH THE CLIMATE CRISIS BECOMING MORE ACUTE, GREEN HYDROGEN COULD BE AN IMPORTANT SOLUTION

N **EARLY ONE-FIFTH OF** global greenhouse gas emissions in 2022 originated from sectors like iron & steel, manufacturing, construction, etc. These sectoral greenhouse gas emissions pose a significant challenge due to technological and financial hurdles

ILLUSTRATION BY **RAJ VERMA**

and are classified as “hard-to-abate” (HTA) sectors.

Current commercially available decarbonisation technologies (renewables, energy efficiency, electrification, etc.) can only achieve 50-60% reductions—closing the gap, therefore, necessitates developing newer solutions. Particularly for HTA sectors, hydrogen has been identified as a key solution. Many of these sectors require chemical feedstocks, or heavy-duty freight solutions, and face significant barriers towards using low-carbon technologies like renewables. Clean hydrogen offers a versatile alternative, capable of direct use in fuel cells, and as feedstock for producing derivatives, making it crucial for addressing limitations of renewables and enabling deep decarbonisation. It is projected that beyond 2030, clean hydrogen demand from HTA sectors could exceed 150 million tonnes per annum (MTPA).

While ‘Clean Hydrogen’ has various production pathways, Green Hydrogen (GH₂), produced through electrolysis using renewable energy (RE), is the most scalable, clean, and safe method and is likely to see the greatest adoption. This is where the opportunity lies for India to play a significant part in the global value chain. India has the fourth largest installed RE (around 5% of global capacity). The RE market in India is cost competitive with tariffs at \$30-50 per Megawatt hour (MWh), lower than countries like Australia, the US, and EU. Setting up RE is the most expensive step in GH₂ production with round-the-clock RE amounting to 50-70% of total project cost. India, therefore, possesses a noticeable cost

